

BEEAWARE

BeeAware Brasil

Product, Data & Engineering Roadmap

Urban Safety Intelligence for Brazil

A complete blueprint covering product strategy, official and journalistic data sources, adapters, normalisation, geospatial architecture, confidence and deduplication engines, wireframes, rollout strategy, data governance, observability and long-term platform evolution.

Version 1.0

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CONFIDENTIAL WORKING BLUEPRINT

Executive Summary

BeeAware Brasil should not be built as a simple country extension. The Brazilian opportunity is to evolve BeeAware from a community incident map into an Urban Safety Intelligence platform capable of combining fragmented public-security data, local journalism, road-safety data and community signals into a single, explainable geographic experience.

Core proposition

BeeAware should answer not only “What happened near me?”, but also “What is happening now?”, “Is this unusual?”, “How reliable is the signal?”, “How does this area compare with its baseline?” and, later, “Which route has fewer recent safety signals?”

Strategic principles

- Evidence over labels: never claim that a place is inherently safe or dangerous.
- Provenance by design: every metric and event must preserve source, date, methodology and geographic precision.
- One product, many adapters: state-specific source logic remains outside the client application.
- Geographic honesty: neighbourhood-level data should be shown as neighbourhood-level data, never converted into invented exact pins.
- Official + contextual + live: combine official data, validated journalism and community reports without conflating them.
- Brazil first, global architecture: design the layer so Spain, Portugal and other markets can later be added through country adapters.

Recommended first-market sequence

Wave	Coverage	Purpose
0	National foundations	SINESP, IBGE, PRF, RENAEST, core schema and adapters
1	Belém + Ananindeua	Neighbourhood intelligence, Pará source integration, local-news pilot
2	São Paulo	Scale, 190 calls, phones/vehicles and large-volume ingestion
3	Rio de Janeiro	CISP/AISP/RISP and geospatial polygon intelligence
4	MG, CE, BA, RS, ES, DF	Expand with high-value structured state sources
5	Remaining capitals / 27 UFs	Complete national state adapter coverage

Roadmap Structure

Section	Topic
1	Product vision and operating principles
2	Reference architecture and data layers
3	National sources
4	State-source registry for 27 UFs
5	Journalistic and community intelligence
6	Canonical taxonomy and data model
7	Geospatial, confidence and deduplication engines
8	API and backend contracts
9	Wireframes and product experience
10	LGPD, moderation and abuse prevention
11	Implementation roadmap and sprint plan
12	Metrics, observability and data quality
13	Long-term platform vision
14	Selected official sources

1. Product Vision and Operating Principles

1.1 From incident map to Urban Safety Intelligence

The Brazilian version should elevate BeeAware from a map that displays incidents into an intelligence layer that explains activity, context, trend, freshness and confidence. A user should be able to understand what the available data says without being pushed into sensational conclusions.

WHAT HAPPENED?	Official + News + Community
WHERE?	PostGIS + official areas + geocoding
WHEN?	Historical / Recent / Live
IS IT UNUSUAL?	Trend engine
CAN I TRUST IT?	Provenance + confidence
DOES IT AFFECT MY WAY?	Route intelligence (later phase)

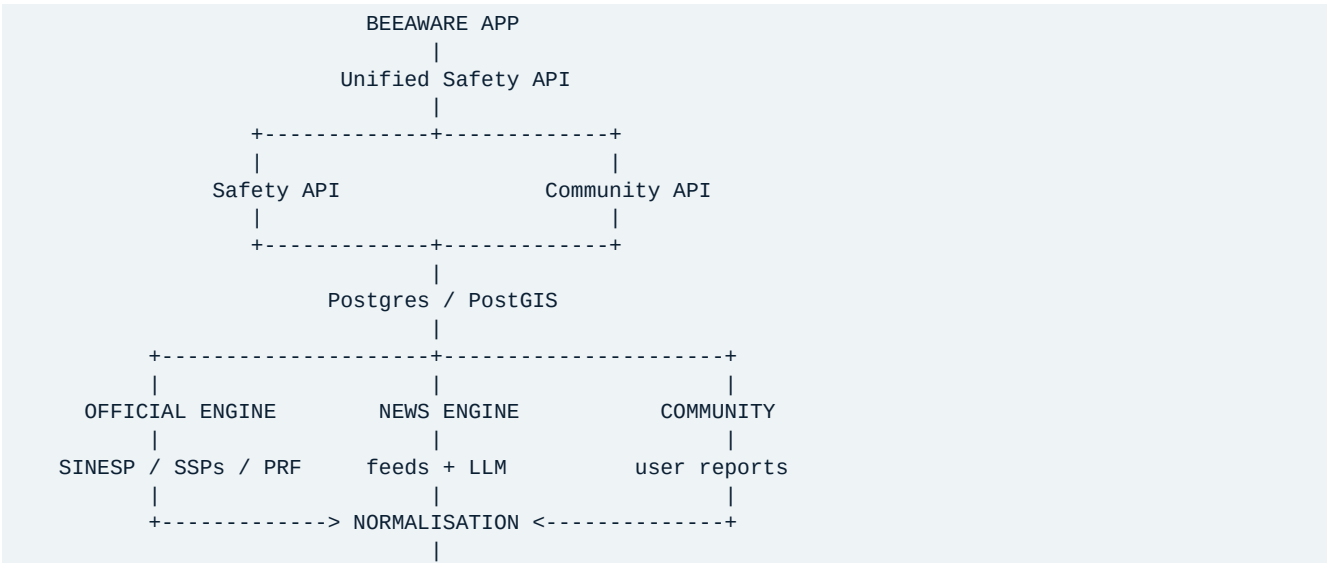
1.2 Three separate safety dimensions

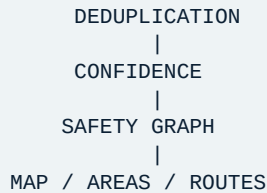
Dimension	Suggested window	Primary purpose
Historical Safety	12 months	Structural baseline based mainly on official statistics, severity and population normalisation.
Recent Activity	30 days	Detect recent changes versus the historical baseline using official, news and community signals.
Live Awareness	24h / 72h / 7 days	Surface current or very recent signals from news, emergency calls, community and road incidents.

Language rule

Do not say “Umarizal is 74% safe”. Say “Historical Safety 74” and explain the components, source coverage and last update. The metric is an intelligence indicator, not a probability of personal safety.

2. Reference Architecture and Data Layers





2.1 Why adapters matter

Brazilian public-security information is fragmented across federal, state and municipal systems. The application must therefore consume a unified BeeAware API, while source-specific parsing, field mapping and update rules live inside adapters.

```

registerAdapter(new PaSegupAdapter());
registerAdapter(new SpSspAdapter());
registerAdapter(new RjIspAdapter());
  
```

The Flutter application should never need to understand that one state publishes CSV, another publishes CKAN resources, another offers spreadsheets, and another exposes geographic polygons. The adapter layer isolates those differences.

3. National Data Sources

3.1 SINESP VDE - national baseline

SINESP is the national backbone for municipal security statistics and the fallback layer for municipalities where BeeAware has not yet implemented a state-level adapter. It should feed historical trend, municipal comparison, baseline metrics and national/state benchmarks. It should not be treated as a source of exact incident pins.

- Coverage: all Brazilian states and the Federal District.
- Use: municipal statistics, historical series, trend and fallback coverage.
- Expected cadence: monitor daily; ingest when a new official release appears.
- Store source-period and methodology metadata because states can revise data and differ methodologically.

Adapter

```
SinespAdapter
```

3.2 IBGE - geographic and population master

IBGE should be the authoritative territorial identity layer. The primary municipality key must be `city_ibge_code`, not free-text city names.

- Municipality geocode and polygon
- UF and region hierarchy
- Population denominator for rates
- Centroids and official territorial boundaries

```
country_code + state_code + city_ibge_code = canonical municipality identity
```

3.3 PRF - federal road safety

PRF open accident data should be part of the MVP because it enables a high-value road-safety layer independent of criminal-occurrence data.

- Historical and current accident datasets
- Road / municipality / date / time / cause / severity
- Road hazard and accident hotspot analysis
- Future support for Route Awareness

Adapter

```
PrfAccidentsAdapter
```

3.4 RENAEST / SENATRAN

RENAEST complements PRF by covering road-traffic records outside the federal-road dataset. The two should remain separate in provenance and be combined only at the intelligence layer.

```
PRF      -> federal highways
RENAEST  -> other available traffic networks
```

Adapter

```
RenaestAdapter
```

3.5 DATASUS / SIM

DATASUS/SIM should be used as an audit and structural-analysis layer rather than as a source of real-time crime pins. It can help test whether mortality trends broadly align with police-record trends and identify potential methodological divergence.

3.6 FBSP / Brazilian Public Security Forum

The Brazilian Public Security Yearbook is valuable as an annual benchmark layer for rankings, cross-checking, national context and research. It should not be used as a primary incident feed.

Adapter

```
FbspBenchmarkAdapter
```

3.7 Atlas da Violência

Use the Atlas as an analytical layer for lethality, demographics and structural trends. Keep it separate from event-level feeds and expose it as context rather than live activity.

4. State Source Registry - 27 UFs

Priority tiers below refer to BeeAware integration readiness and strategic value, not to whether a state has “good” or “bad” security data. Tier C means SINESP should be used initially until the state-specific publication method has been fully validated for production automation.

UF	Primary source	Planned adapter	Tier
AC	SSP-AC + SINESP	AcSecurityAdapter	C
AL	SSP-AL	AlSecurityAdapter	B/C
AP	SEJUSP-AP	ApSecurityAdapter	C
AM	SSP-AM	AmSecurityAdapter	B
BA	SSP-BA	BaSspAdapter	A
CE	SSPDS / SUPESP	CeSupespAdapter	A
DF	SSP-DF	DfSspAdapter	A/B
ES	SESP-ES	EsSespAdapter	A
GO	SSP-GO	GoSspAdapter	B/C
MA	SSP-MA	MaSspAdapter	B/C
MT	SESP-MT	MtSespAdapter	B/C
MS	SEJUSP-MS	MsSejuspAdapter	B/C
MG	Dados MG / SEJUSP	MgSejuspAdapter	A+
PA	SEGUP / CODEC / SIAC	PaSegupAdapter	A+
PB	SESDS-PB	PbSedsAdapter	B/C
PR	SESP-PR	PrSespAdapter	B
PE	SDS-PE	PeSdsAdapter	B
PI	SSP-PI	PiSspAdapter	C
RJ	ISP	RjIspAdapter	A+
RN	SESED-RN	RnSesedAdapter	C
RS	SSP-RS	RsSspAdapter	A+
RO	SESDEC-RO	RoSesdecAdapter	C
RR	SESP-RR	RrSespAdapter	C
SC	SSP-SC	ScSspAdapter	B
SP	SSP-SP	SpSspAdapter	A+
SE	SSP-SE / CEACRIM	SeSspAdapter	B
TO	SSP-TO	ToSspAdapter	C

4.1 Pará - SEGUP / CODEC / SIAC

Pará is an unusually strong pilot candidate because the state portal allows analysis by municipality, neighbourhood, offence type and period. This supports the type of local context that BeeAware needs for Belém and Ananindeua.

- Target cities: Belém, Ananindeua and later Marituba.
- Primary value: neighbourhood-level statistics.
- Use both original source category and BeeAware-normalised category.
- Build export/ingestion checks that validate changes in source structure.

Adapter

PaSegupAdapter

4.2 São Paulo - SSP-SP

São Paulo should test scale and source diversity. Separate adapters are recommended for general crime statistics, emergency calls, phone-related events, vehicle data and IML data.

```
SpCrimeAdapter
Sp190Adapter
SpPhoneAdapter
SpVehicleAdapter
SpImlAdapter
```

Important distinction

A 190 emergency call and a police occurrence report are not the same event type. They can corroborate one another but should never be counted as interchangeable observations.

4.3 Rio de Janeiro - ISP

Rio should be the flagship spatial-intelligence implementation because ISP provides the CISP/AISP/RISP hierarchy and geographic boundary files. These polygons should be stored in PostGIS and linked to official statistics by version and effective date.

```
user GPS
  |
  ST_Contains()
  |
  CISP -> AISP -> RISP -> matching statistics
```

Adapters

RjIspAdapter
RjGeoAdapter

4.4 Minas Gerais

Prioritise machine-readable resources and CKAN/DataStore access where available. Minas is well suited to automated ingestion and can become a model for clean state adapters.

Adapter

MgSejuspAdapter

4.5 Rio Grande do Sul

Use a file-ingestion pipeline with checksum and schema validation for regularly published microdata packages.

```
check source -> checksum -> download -> unzip -> validate schema -> normalize -> upsert
```


Adapter

RsSspAdapter

4.6 Bahia

Use state microdata and territorial groupings such as RISP/AISP where available. Preserve original offence terminology and personal-data minimisation rules before any event reaches the public layer.

Adapter

BaSspAdapter

4.7 Ceará

Ceará is a high-priority second-wave state. Build category-specific mappings for CVLI, CVP, theft, sexual crimes, domestic violence, weapons and drugs, with support for integrated security areas.

Adapter

CeSupespAdapter

4.8 Espírito Santo

Prioritise structured datasets for homicide/femicide, phone robbery, domestic violence, cybercrime and fraud.

Adapter

EsSespAdapter

5. Journalistic and Community Intelligence

5.1 BeeAware News Intelligence

News sources provide recency that monthly official statistics often cannot. They should be treated as a separate signal class, with source reputation, extraction confidence and deduplication before public display.

- National and state G1 editions
- CNN Brasil, Band, Record, UOL and other established newsrooms
- State and municipal newspapers
- Hyperlocal outlets where editorial standards are acceptable
- Official SSP, Civil Police, Military Police, Fire Brigade, Civil Defence and traffic feeds

```
feed/source -> fetch -> clean -> classify -> incident? -> extract location/time/type -> geocode
-> dedupe -> confidence -> event
```

5.2 Incident vs context classification

```
Example A: "Phone stolen near a shopping centre in Umarizal tonight"
=> is_event=true, category=PROPERTY, type=phone_robbery
```

```
Example B: "Robbery falls 18% in Pará in 2026"
=> is_event=false, content_type=statistical_context
```

5.3 Community reports

Community reporting should remain a core BeeAware differentiator but must visibly distinguish unverified reports from official or independently corroborated events.

```
report -> automated validation -> abuse/spam check -> location validation -> publish UNVERIFIED
-> corroboration -> verified-source match -> confidence update
```

Suggested statuses

```
reported -> corroborated -> verified -> resolved -> expired
```

6. Canonical Taxonomy and Data Model

6.1 Universal taxonomy

```
VIOLENCE
  homicide
  attempted_homicide
  assault
  sexual_violence
  domestic_violence
  police_intervention
  kidnapping

PROPERTY
  robbery
  theft
  burglary
  phone_robbery
```

```

phone_theft
vehicle_robbery
vehicle_theft
cargo_robbery

PUBLIC_SAFETY
weapon
drugs
disturbance
suspicious_activity
fire
emergency

ROAD_SAFETY
accident
serious_accident
fatal_accident
road_hazard
road_closure

COMMUNITY
suspicious_activity
harassment
disorder
unsafe_environment
other

```

Always preserve both `original_category` and `normalized_category`. This supports auditability, state-by-state mapping changes and future reprocessing without loss of source semantics.

6.2 security_events

```

security_events
  id uuid
  country_code
  state_code
  city_ibge_code
  source_id
  source_record_id
  source_type
  event_category
  event_type
  event_subtype
  original_category
  occurred_at
  reported_at
  published_at
  latitude / longitude / geometry
  geo_precision
  location_confidence
  neighborhood / district / city / state
  occurrence_count / victim_count
  severity
  confidence_score
  official / community / news
  canonical_event_id
  raw_payload jsonb
  created_at / updated_at

```

6.3 security_sources

```

security_sources

```

```
id
country / state
name / organisation
source_type / source_url
adapter_name / adapter_version
refresh_frequency
last_check / last_success / last_data_date
geo_precision
reliability_score / quality_score
terms_reviewed_at
active
```

6.4 geo_areas

```
geo_areas
  id
  country / state / city_ibge_code
  area_type
  name
  geometry
  parent_area_id
  source / source_version
  valid_from / valid_to
```

Area types should support COUNTRY, STATE, MUNICIPALITY, NEIGHBORHOOD, RISP, AISP, CISP, AIS, RA and other official territorial constructs.

7. Geospatial, Confidence and Deduplication Engines

7.1 PostGIS as a core dependency

```
ST_DWithin()
ST_Contains()
ST_Intersects()
ST_Distance()
ST_ClusterDBSCAN()
```

These operations power nearby incidents, hotspot analysis, official police-area mapping, radius search, route buffers and geographic clustering.

7.2 Geographic precision

Precision	Display behaviour
EXACT	Use exact or near-exact location where the source legitimately provides it.
STREET	Display a street-level area; avoid house-level inference.
NEIGHBORHOOD	Use neighbourhood polygon or aggregated cluster.
DISTRICT	Display district-level context.
MUNICIPALITY	City-level statistic only; no invented pin.
STATE	State-level context only.

7.3 Confidence engine

Signal type	Starting reliability
Official confirmed record	1.00
Official emergency call	0.90
Major media + precise location	0.85
Established local journalism	0.75
Multiple independent reports	0.70
Verified community member	0.55
Single anonymous report	0.30
AI inference alone	0.00 public

```
confidence = source_reliability × location_confidence × time_confidence ×
corroboration_adjustment
```

AI rule

An AI-only inference must never be displayed to users as an event. AI is a classifier/extractor, not a factual source.

7.4 Deduplication engine

Multiple articles, a police release and a community report may refer to the same occurrence. BeeAware should build one canonical event with multiple supporting source records.

```
canonical_event
| - source_event_G1
| - source_event_UOL
| - source_event_SSP
` - community_report
```

- Candidate matching: time window
- Geographic distance / polygon overlap
- Normalised category
- Named entities
- Semantic similarity
- Source provenance and corroboration

8. API and Backend Contracts

8.1 Suggested backend structure

```
/src
  /adapters
    /global
    /br
      sinesp/ ibge/ prf/ renaest/ datasus/ fbsp/
    /states
      pa/ sp/ rj/ mg/ rs/ ba/ ce/ es/ pr/ pe/ df/ ...
  /services
    geocoding/ classification/ normalization/ deduplication/
    confidence/ trends/ safety-pulse/ source-health/
  /api
    nearby/ areas/ events/ routes/ reports/ sources/
```

8.2 Adapter contract

```
interface SecuritySourceAdapter {
  source(): SecuritySource;
  fetch(since?: Date): Promise<RawSecurityRecord[]>;
  normalize(record: RawSecurityRecord): Promise<SecurityEvent[]>;
  healthCheck(): Promise<SourceHealth>;
}
```

8.3 Unified endpoints

```
GET /v1/security/nearby
GET /v1/security/areas/{id}
GET /v1/security/areas/{id}/trends
GET /v1/security/events
GET /v1/security/road-awareness
GET /v1/security/sources
POST /v1/security/community-report
```

8.4 Example nearby response

```
{
  "location": {"country": "BR", "state": "PA", "city": "Belém", "neighborhood": "Umarizal"},
  "pulse": {"historical": 74, "recent": 61, "live": 83},
  "events": [],
  "trends": [],
  "source_quality": 0.88,
  "updated_at": "2026-08-21T00:00:00Z"
}
```

9. Wireframes and Product Experience

9.1 Home / Map

```
+-----+
| BeeAware                               [bell] [u] |
| [ Search place ]                         |
+-----+
|
|      MAP
|    o    !    o
|      *
|    [you]
|
+-----+
| Umarizal                               |
| Live Awareness             78          |
| 3 recent signals           |
| activity above normal      |
+-----+
| MAP  AREA  +REPORT  ROUTES            |
+-----+
```

9.2 Area Intelligence

```
+-----+
| < Umarizal                             |
|   Belém, Pará                          |
+-----+
| SAFETY PULSE                           |
| Historical   74                         |
| Recent      61                         |
| Live        83                         |
+-----+
| Last 30 days                            |
| Robbery      +12%                      |
| Theft         2%                      |
| Vehicles      - 8%                    |
| Violence      - 4%                    |
+-----+
| Higher activity: 18:00-22:00            |
| Nearby: Nazaré 81 / Marco 73           |
+-----+
| Sources: SEGUP | SINESP | News | BA   |
+-----+
```

9.3 Event Detail

```
+-----+
| < Event                                 |
+-----+
| Phone robbery                           |
| Umarizal - Belém                       |
| Today - approx. 18:40                   |
| Geographic precision: neighbourhood    |
| Confirmed by 2 sources                  |
| [news] [community]                     |
| No indication of an active threat.     |
+-----+
| View original information >            |
+-----+
```



```
| Report incorrect information |
+-----+
```

9.4 Community Report

```
+-----+
| REPORT |
+-----+
| What happened? |
| ( ) robbery / theft |
| ( ) suspicious activity |
| ( ) harassment |
| ( ) accident |
| ( ) disturbance |
| ( ) other |
| |
| Where? [use my location] |
| When? [now] |
| Optional details |
| [ Send anonymously ] |
+-----+
```

9.5 Route Awareness - later phase

```
+-----+
| ROUTE AWARENESS |
+-----+
| From: My location |
| To: School |
+-----+
| Route A      82      28 min |
| 1 attention point |
| |
| Route B      71      25 min |
| 3 attention points |
| |
| Faster: B |
| Fewer recent safety signals: A |
+-----+
```

Naming rule

Avoid “Safest Route”. Use “Route Awareness” or “Route with fewer recent safety signals”. This preserves methodological accuracy and reduces the risk of implying a guarantee of safety.

10. LGPD, Moderation and Abuse Prevention

10.1 LGPD by design

BeeAware should minimise personal data at ingestion, processing and public display. The product is about environmental awareness, not building a searchable catalogue of people involved in incidents.

- Do not publicly store/display victim names, CPF, phone numbers, identifiable licence plates, residential addresses, facial imagery or identifiable data about minors unless strictly necessary and legally justified.
- Apply purpose limitation, data minimisation, retention limits, role-based access and audit logs.
- Strip unnecessary personal fields from state microdata before placing records in the public data layer.

10.2 Abuse prevention

- Block doxxing and accusations naming individuals.
- Prevent reports designed to follow or locate a person.
- Do not support retaliatory tracking or harassment.
- Community reports describe situation, place and time - not identities of alleged suspects.
- Use rate limits, duplicate checks, device/account reputation and moderation queues.

11. Implementation Roadmap and Sprint Plan

11.1 Phase 0 - Foundation

- Create country-aware architecture
- Create security_sources registry
- Create security_events v2
- Create geo_areas and PostGIS extension
- Define universal taxonomy
- Define adapter framework
- Define provenance, confidence and geographic precision models

Go / No-Go

The schema must accept UK and Brazil data without country-specific columns or client-side special cases.

11.2 Phase 1 - National Brazil Layer

- IBGE adapter
- SINESP adapter
- PRF adapter
- RENAEST adapter
- National municipality pages
- National road-safety baseline

Outcome: BeeAware can display baseline information for the whole country even before state adapters are completed.

11.3 Phase 2 - Belém pilot

- SEGUP-PA adapter
- Neighbourhood polygons / mappings
- Belém and Ananindeua area pages
- Local-news source set
- Community reports with Brazilian taxonomy
- Initial data-quality dashboard

Purpose: validate neighbourhood intelligence and a local-news pipeline in a market with meaningful crime data and strong local relevance.

11.4 Phase 3 - São Paulo

- General SSP statistics
- 190 calls
- Phone data
- Vehicle data
- IML where appropriate
- High-volume ingestion/load testing

Purpose: validate scale, multiple source types and deduplication under high event volume.

11.5 Phase 4 - Rio de Janeiro

- ISP statistics
- CISP/AISP/RISP polygons
- Spatial joins and area intelligence

- Versioned territorial mapping

Purpose: validate the geographic intelligence layer and official security-area hierarchy.

11.6 Phase 5 - Safety Pulse

- Historical Safety
- Recent Activity
- Live Awareness
- Population normalisation
- Outlier handling
- Coverage adjustment
- Source-quality exposure

Release gate

Do not release a user-facing score until normalisation, coverage bias and source freshness have been validated in the first three pilot markets.

11.7 Phase 6 - News Intelligence

- Configurable feed registry
- Article classifier
- Location and time extraction
- Geocoder
- Source-reputation model
- Deduplication to canonical events

11.8 Phase 7 - Community Intelligence 2.0

- Report corroboration
- Reputation / trust signals
- Duplicate detection
- Report ageing and expiry
- Status model
- Source matching

11.9 Phase 8 - Second-wave states

Recommended order: Minas Gerais, Ceará, Bahia, Rio Grande do Sul, Espírito Santo and Distrito Federal. This provides a mix of structured sources, regional diversity and large urban populations.

11.10 Phase 9 - Remaining capitals

Expand to Recife, Curitiba, Florianópolis, Manaus, São Luís, João Pessoa, Natal, Maceió, Teresina, Goiânia, Campo Grande, Cuiabá, Porto Velho, Rio Branco, Macapá, Boa Vista, Palmas and Aracaju, while progressively improving state adapters beyond the capitals.

11.11 Phase 10 - 27/27 state adapters

Goal: complete state-level adapter coverage while retaining SINESP as the national reference and consistency layer.

11.12 Phase 11 - Route Awareness

route geometry -> 250-500m buffer -> recent events + historical activity + road hazards ->

The product should compare routes by recent safety signals rather than claiming that one route is guaranteed to be safer.

11.13 Phase 12 - Alerts

- Home
- Work
- School
- Current area
- Custom places

Prioritise meaningful changes, not notification spam. A useful alert is “unusual increase in robbery signals over the last seven days near Home”, not a push notification for every individual occurrence.

11.14 Phase 13 - Personal context

Allow non-sensitive journey context such as walking, driving, cycling, public transport and daypart preferences. Do not build sensitive personal-risk profiles.

11.15 Phase 14 - Predictive insights

Only introduce historically grounded patterns, such as “records in this area tend to be higher between 18:00 and 22:00”. Avoid event prediction or language suggesting that a crime will occur.

11.16 Suggested sprint sequence

Sprint	Primary delivery
1	Schema + source registry
2	PostGIS + geo areas
3	Adapter framework
4	IBGE adapter
5	SINESP adapter
6	National statistics
7	PRF
8	SEGUP-PA
9	Belém UI
10	News engine MVP
11	Confidence + provenance
12	Deduplication
13	SSP-SP
14	São Paulo 190
15	ISP-RJ
16	CISP/AISP GIS
17	Safety Pulse
18	Community v2
19	MG / CE / BA

20	RS / ES / DF
21+	Remaining states and capabilities

11.17 MVP Brazil definition

Dimension	MVP scope
Markets	Belém, São Paulo and Rio de Janeiro
Core official data	SINESP, IBGE, PRF, SEGUP-PA, SSP-SP, ISP-RJ
Context signals	Selected news feeds and community reports
Product	Map, nearby, area page, event details, filters, report, source transparency, trends
Not yet	Route intelligence, advanced prediction, all-state adapters, complex alerting

12. Metrics, Observability and Data Quality

12.1 North-star metric

Weekly Informed Users

Users who meaningfully consult safety information about an area, a followed place or a route during the week. This is a stronger behavioural measure than simply counting incident views.

12.2 Product metrics

- WAU / MAU
- Locations followed
- Reports submitted
- Reports corroborated
- Event confirmation rate
- Return rate
- Alert open rate
- Coverage by city/neighbourhood
- Data freshness seen by users

12.3 Technical metrics

- Adapter availability
- Ingestion latency
- Data freshness
- Failed records
- Unknown categories
- Geocoding failures
- Duplicates detected
- Classification confidence
- Source coverage
- Schema-change alerts

12.4 Data Quality Engine

Each source should receive explicit quality dimensions rather than one opaque health flag.

Dimension	Meaning
Freshness	How recent is the latest published period?
Completeness	Are expected rows/areas/categories present?
Consistency	Does the schema and methodology remain stable?
Geo precision	How granular is the location?
Historical continuity	Are there breaks or missing periods?
Methodological stability	Have definitions or reporting rules changed?

12.5 Source Health Dashboard

SOURCE	LAST DATA	STATUS
SINESP	Jul-26	GREEN
SEGUP-PA	Aug-26	GREEN

ISP-RJ	Jul-26	GREEN
SSP-SP	Jul-26	GREEN
PRF	Jul-26	GREEN
RENAEST	Mar-26	AMBER
SSP-XX	Apr-26	RED

12.6 Jobs and update checks

Process	Check cadence
News	Every 15 minutes
Community	Real time
Source health	Daily
State datasets	Daily check
SINESP	Daily check
PRF	Daily check
RENAEST	Weekly check
FBSP	Monthly check
IBGE	Monthly check

Where possible store ETag, Last-Modified, checksum, file size and latest available period to avoid needless reprocessing.

```
source check -> changed? -> no: stop / yes: download -> validate -> ingest
```

13. Long-Term Platform Vision

13.1 The strategic moat

The long-term defensibility is not the map itself. It is the BeeAware Safety Data Graph: a system able to reconcile fragmented safety sources into explainable, source-aware, geographically honest intelligence.

```

BeeAware UK
|
BeeAware Brazil
|
BeeAware Spain
|
BeeAware Portugal
|
v
BeeAware Global Safety Graph

```

13.2 Expansion rule

Adding a new city should eventually mean registering or extending an adapter, not rewriting the product. Once a source enters the shared pipeline, normalisation, geocoding, deduplication, confidence scoring, area aggregation and API presentation should be reusable.

```
new state source -> adapter -> normalize -> geocode -> dedupe -> confidence -> BeeAware
```


13.3 Editorial tone

- Calm
- Objective
- Transparent
- Non-sensationalist
- Explain uncertainty
- Avoid causal claims that the data does not support

Preferred wording

“More robbery records were observed than the recent average for this area.” Avoid language such as “area dominated by crime” or “extremely dangerous region”.

13.4 First true milestone

BeeAware Brazil Data Layer v1

The first milestone is reached when GET /v1/security/nearby for a coordinate in Belém can consistently return municipality/neighbourhood context, IBGE geography, SINESP trend, SEGUP statistics, PRF proximity where relevant, recent news, community reports, provenance and confidence - without Flutter needing to know which sources produced the result.

14. Selected Official and Reference Sources

These are the principal public sources that informed the roadmap. Source structures and publication methods may change, so adapters should maintain terms-review dates, health checks and schema-change detection.

Source	Role	URL
MJSP / SINESP VDE	National public-security statistics and historical bases	https://www.gov.br/mj/pt-br/assuntos/sua-seguranca/seguranca-publica/estatistica
MJSP - Mapa da Segurança Pública	National indicators and annual overview	https://www.gov.br/mj/pt-br/assuntos/noticias-1/mapa-da-seguranca-publica-2026-traz-30-indicadores-nacionais-sobre-criminalidade
IBGE Malhas Territoriais	Municipality geocodes and geographic boundaries	https://www.ibge.gov.br/geociencias/organizacao-do-territorio/malhas-territoriais/15774-malhas.html
PRF Dados Abertos	Federal-road accident datasets	https://www.gov.br/prf/pt-br/aceso-a-informacao/dados-abertos/dados-abertos-da-prf
SENATRAN / RENAEST	National traffic accident statistics	https://www.gov.br/transportes/pt-br/assuntos/transito/conteudo-Senatran/registro-nacional-de-sinistros-e-estatisticas-de-transito
SEGUP Pará / CODEC	Pará security statistics with municipality/neighbourhood filters	https://codec.segup.pa.gov.br/
SSP São Paulo	Crime statistics and open datasets including 190 calls	https://www.ssp.sp.gov.br/estatistica/consultas
ISP Dados - Rio de Janeiro	RJ statistics, CISP/AISP/RISP and geographic files	https://www.ispdados.rj.gov.br/
Dados MG	Structured Minas Gerais open datasets	https://www.dados.mg.gov.br/
SSP Rio Grande do Sul	State microdata and open security data	https://www.ssp.rs.gov.br/dados-abertos
SSP Bahia	State open security data and microdata documentation	https://www.ba.gov.br/ssp/dados-abertos
SSPDS / SUPESP Ceará	Security indicators and territorial statistics	https://www.ce.gov.br/sspds/estatisticas/
SESP Espírito Santo	State security datasets	https://dados.es.gov.br/organization/sesp-secretaria-de-estado-da-seguranca-publica-e-defesa-social
Fórum Brasileiro de Segurança Pública	Brazilian Public Security Yearbook	https://forumseguranca.org.br/publicacoes/anuario-brasileiro-de-seguranca-publica/

Closing principle

Build the data layer first. Once BeeAware can reliably explain what happened, where, when, according to which sources and with what confidence, the interface becomes a presentation layer over a genuinely scalable safety-intelligence platform.